Sagar Varma

Computer Engineering student @ UCSB building AI agents, dev infra, and real-time systems

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EXPERIENCE

SemiAnalysis Oct 2025 – Sep 2026

Machine Learning Engineering Intern, Remote

- Built deep research AI agents that mapped thousands of semiconductor fabs at site level, saving hundreds of research hours
- Researched fabs by hand to build ground-truth data, then automated it into LLM evaluations (evals)
- Designed verifier agents that catch hallucinations before they reach the dataset
- Built internal AI tools that researchers across the company used to query and visualize research data

Avora AI (Y Combinator) Jun 2025 – Sep 2025

Software Engineering Intern, San Francisco, CA

- Scaled the FastAPI backend to 1,000+ dental practices using the product concurrently
- Shipped practice telemetry and quality-of-life features that raised retention 55%

Technologies: Python, TypeScript, React, Remix, PostgreSQL, FastAPI, SQLAlchemy, Kubernetes, GCP BigQuery, AssemblyAI

Intel Jun 2022 – Sep 2022

Software Engineering Intern, Folsom, CA

- Automated manual code updates across millions of lines, cutting merge conflicts 99%
- Shipped security patches for legacy graphics drivers

Technologies: Python, C, Perl, Ruby

PROJECTS

Orbit | orbitdeploy.dev | github.com/sagarvarma1/orbit

Open-source, self-hostable Vercel alternative that deploys apps from Git

- Turns a GitHub repo into a live website on every push, the same workflow as Vercel but on any Linux server
- Built Next.js SSR with streaming, branch previews, instant rollbacks, auto-HTTPS custom domains and vercel.json support
- Hosts my other projects in production and is open to new users

Leaf | leaf.orbitdeploy.dev | github.com/sagarvarma1/leaf

Notion-style workspace with blocks, nested pages, real-time collaboration and offline sync

- Lets people write docs and wikis together like Notion, with edits showing up live for everyone and working offline
- Wrote and open-sourced my own conflict-resolution sync engine that merges multiplayer and offline edits; deployed on Orbit

Bardie | textbardie.com

AI assistant that texts college students before they miss deadlines

- Built the agent harness and Canvas LMS connectors; launched at UCSB with a few dozen users and first paying Pro subscribers

Appetite Pal: GLP-1 Food Log | iOS App Store

AI food log for people on GLP-1 medications

- Built and shipped the iOS app; AI tracks calories and every macronutrient
- Sells subscriptions at $14.99/month or $169.99/year

Earlier projects: americantimeuse.com, wheretowatch.fun, Wherely (~1,500 MAU), AI Test Prep Academy (~1,000 waitlist)

TECHNICAL SKILLS

AI Engineering: Large language models (LLMs), generative AI, agentic AI, multi-agent systems, agent harnesses, tool calling, Model Context Protocol (MCP), LLM evals, LLM-as-a-judge, retrieval-augmented generation (RAG), context and prompt engineering

Machine Learning: PyTorch, TensorFlow, fine-tuning, computer vision, natural language processing (NLP)

Frameworks: Claude Agent SDK, OpenAI Agents SDK, LangGraph, LangChain, LlamaIndex, Claude and OpenAI APIs, Claude Code

Languages: Python, TypeScript, JavaScript, Swift, Java, C, C++, SQL

Web & Mobile: FastAPI, Django, Flask, Node.js, React (React.js), Remix, SQLAlchemy, Pydantic, REST APIs, OAuth 2.0, SwiftUI (iOS)

Infrastructure: PostgreSQL, Redis, vector databases, BigQuery, Docker, Kubernetes, AWS, GCP, GitHub Actions (CI/CD)

Hardware: Verilog, VLSI design, Cadence Virtuoso, Cadence SKILL, electronic design automation (EDA), computer architecture

EDUCATION

University of California, Santa Barbara (UCSB) Expected Jun 2027

Bachelor of Science (B.S.), Computer Engineering

- Coursework: AI in Chip Design, VLSI Design, Computer Architecture, Compilers, Semiconductor Fabrication Technology

ADDITIONAL EXPERIENCE

Independent AI Consulting, agents and internal tools for companies (launching) 2026 – Present

Perplexity, Campus Growth Ambassador at UCSB; drove ~500 signups Sep 2025 – Dec 2025